

RAJALAKSHMI
ENGINEERING COLLEGE
An AUTONOMOUS Institution
Affiliated to ANNA UNIVERSITY, Chennai

Department of Cyber Security

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III Year /V Sem

CS23533-FOUNDATIONS OF ARTIFICIAL INTELLIGENCE

Review 1– 25.09.2026

Introduction

- CyberTrace AI is an AI-powered digital forensics platform.
- It analyzes logs from EVTX, CSV, JSON, XML and Linux systems.
- It helps investigators build timelines and detect suspicious activity.
- It generates AI-assisted investigation reports.

Problem Statement

- Cyber incidents produce large volumes of different log files.
- Manual log analysis is slow and difficult.
- Correlating events from multiple sources is challenging.
- A unified system is needed for faster evidence analysis and reporting.

Motivation

- Why was this project selected?:

  Security teams need faster, repeatable incident investigation.
- Research / industry motivation:

  Modern incidents leave evidence across endpoints, servers and application logs
- Expected benefits:

  Automated normalization and correlation reduce repetitive analyst work.
- Practical significance:

  Evidence hashes, audit records and reports support accountable investigations.

Objectives

- Ingest and verify digital evidence using SHA-256 hashes.
- Parse EVTX, LOG, CSV, JSON and XML records into a common schema.
- Correlate related events and reconstruct chronological attack timelines.
- Detect suspicious activity and present it through dashboards and graphs.
- Generate AI-assisted investigation summaries and downloadable reports.

Literature Survey

S.No.	Author & Year	Method / Technique	Key Findings	Limitations
1	Kent & Souppaya, 2006	NIST SP 800-92 log management	Structured log management supports security analysis and incident investigation.	Planning guidance, not an end-to-end investigation system.
2	Kent et al., 2006	NIST SP 800-86 digital forensics	Forensics follows collection, examination, analysis and reporting stages.	Does not prescribe an automated AI workflow.
3	MITRE ATT&CK, 2026	Adversary behavior knowledge base	Techniques provide a common vocabulary for detection and investigation.	Requires evidence mapping and analyst context.

Existing System

Existing approach

Manual log review with separate tools and format-specific analysis.

Advantages

Specialized tools can inspect individual evidence sources.

Limitations

Cross-source correlation, timeline creation and reporting remain analyst-intensive.

Proposed System

Proposed approach

A unified case workflow ingests, verifies, parses and normalizes evidence.

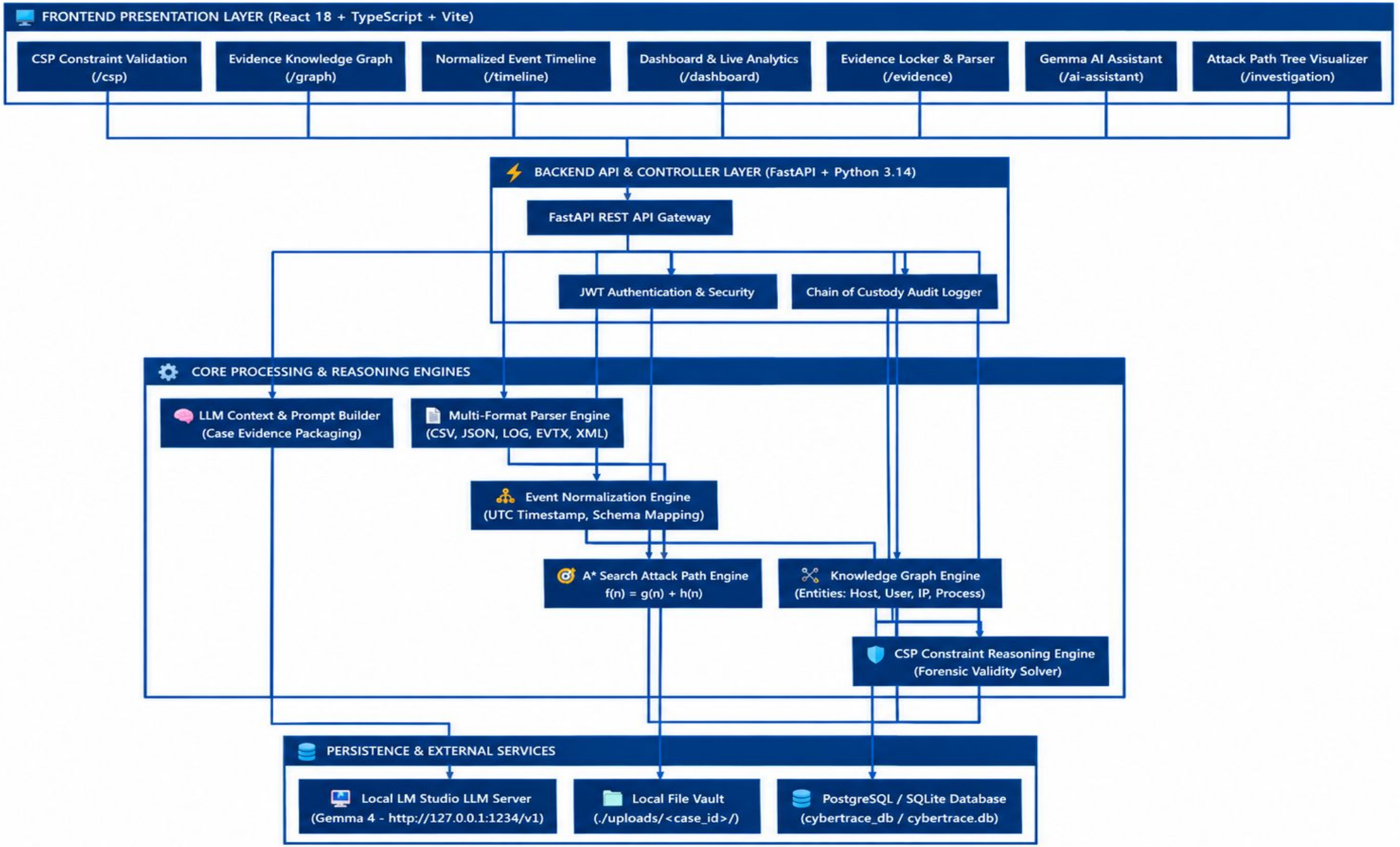
Key capabilities

Correlation, timeline reconstruction, threat detection, knowledge graphs and AI assistance.

Expected outcome

A traceable investigation view with exportable reports.

System Architecture / Block Diagram



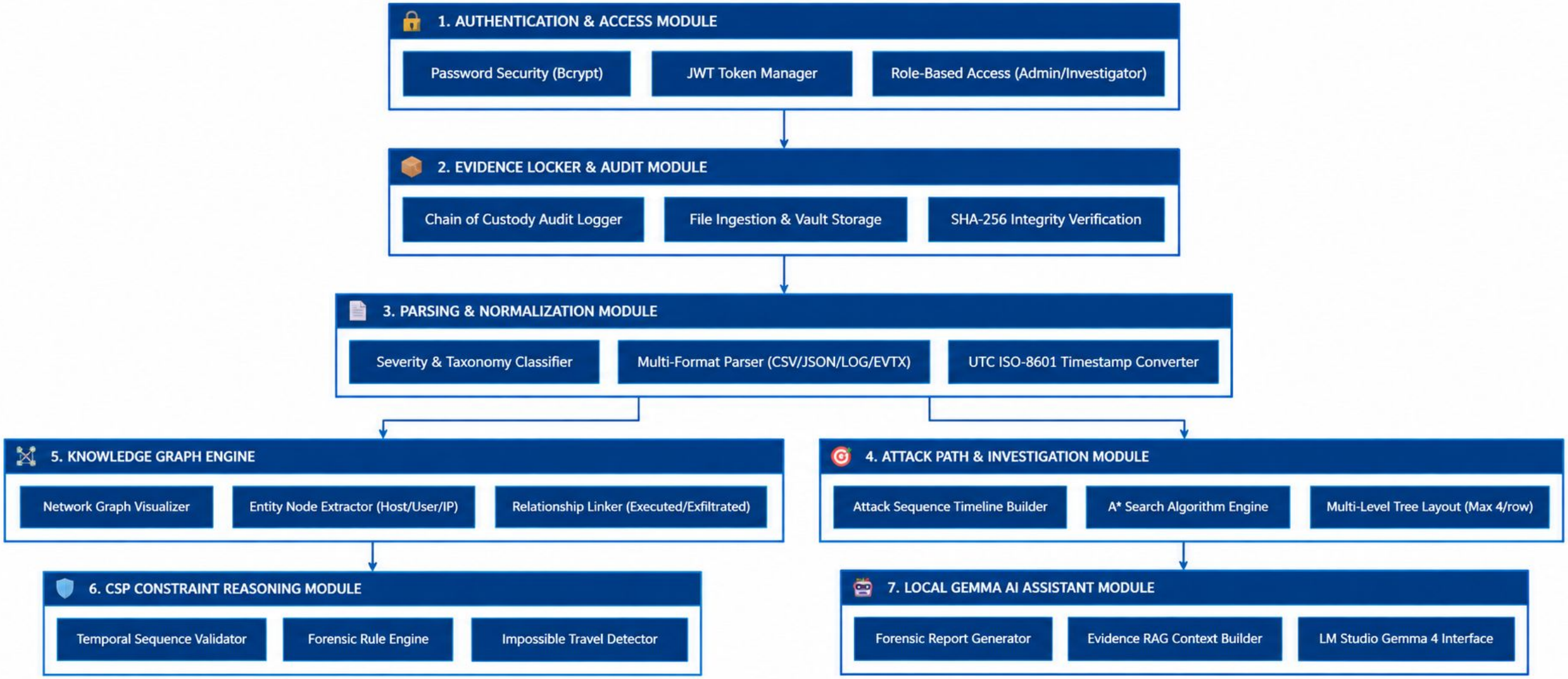
Methodology



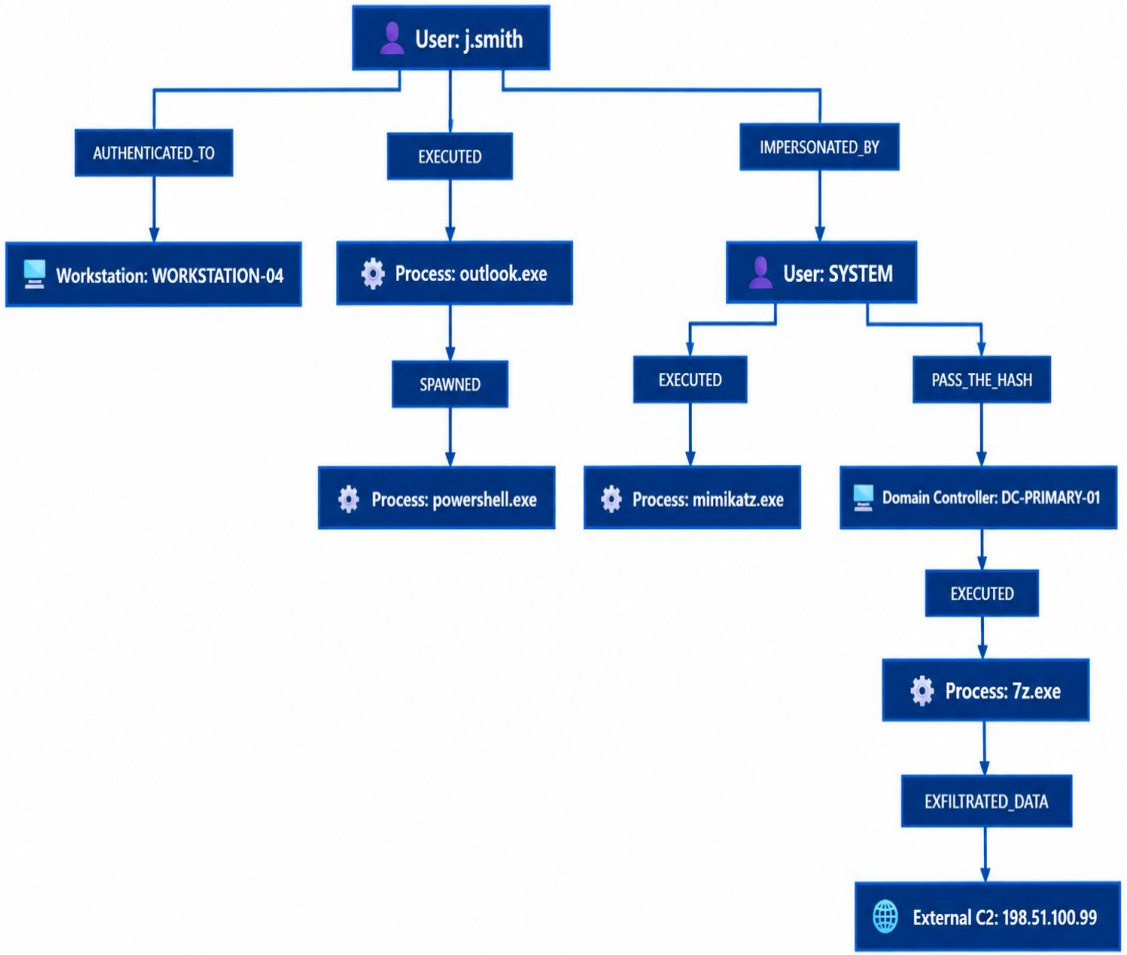
Technologies Used

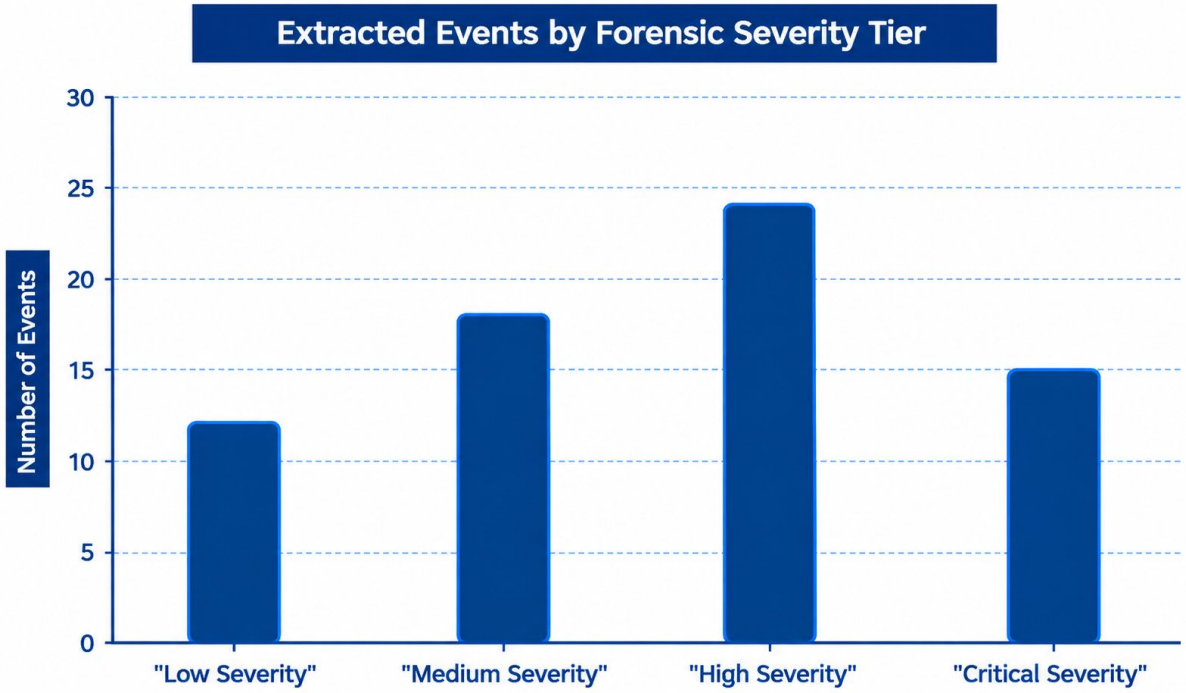
- Frontend: React 19, TypeScript, Vite, Tailwind CSS, Chart.js and React Flow
- Backend: Python 3.12+, FastAPI, SQLAlchemy, Pydantic and Uvicorn
- Database & cache: PostgreSQL 15 and Redis 7
- AI / search: OpenAI API, LangChain, FAISS and Sentence Transformers
- Deployment & tools: Docker Compose, Git, REST APIs and pytest
- Recommended development system: 8-core CPU, 16 GB RAM and 100 GB SSD

Module Description



Preliminary / Expected Results





Project Schedule

Activity	Month 1	Month 2	Month 3	Month 4
Planning & requirements	Completed			
UI & database design	Completed	Completed		
Core implementation		In progress	In progress	
Parser, correlation & AI			Planned	In progress
Testing, reports & documentation				Planned

Current Progress

- ✓ Literature survey completed
- ✓ Problem identification completed
- ✓ Requirement analysis / methodology finalized
- ✓ Implementation completed
- ✓ Testing completed
- ☐ Final evaluation completed

Future Work

- Complete end-to-end testing with varied evidence sets.
- Measure parsing quality, timeline accuracy and response time against stated targets.
- Add MITRE ATT&CK mapping, Sigma/YARA rules and PCAP analysis.
- Strengthen evidence-grounded AI prompts and evaluate analyst usefulness.
- Finalize deployment, user documentation and investigation report templates.

Conclusion

- CyberTrace AI unifies evidence handling, multi-format log parsing, normalization, correlation and timeline reconstruction in one case-centric platform.
- Its core contribution is a traceable investigation workflow that connects raw evidence to alerts, graphs, AI-assisted analysis and downloadable reports.
- The system aims to reduce manual investigation effort while retaining evidence integrity and analyst oversight.

References

[1] K. Kent and M. Souppaya, Guide to Computer Security Log Management, NIST SP 800-92, 2006.

[2] K. Kent et al., Guide to Integrating Forensic Techniques into Incident Response, NIST SP 800-86, 2006.

[3] MITRE, ATT&CK® Enterprise Knowledge Base, accessed September 2026.

[4] CyberTrace AI Project Charter, PRD, TRD, HLD and source repository, 2026.